

Class Test 2 Answer Key

Q1. In fruit flies (*Drosophila melanogaster*), wing shape is controlled by a single gene where normal wings (V) are dominant to vestigial wings (v). Body colour is controlled by a separate gene where gray body (E) is dominant to ebony body (e). A true-breeding normal-winged, gray-bodied fly is crossed with a true-breeding vestigial-winged, ebony-bodied fly to produce the F1 generation.

a) State the genotype and phenotype of all F1 offspring.

Genotype: VvEe

Phenotype: Normal wings, gray body

b) Two F1 flies are crossed. State the expected phenotypic ratio in the F2 generation.

9 normal wings, gray body : 3 normal wings, ebony body : 3 vestigial wings, gray body : 1 vestigial wings, ebony body

c) What important assumption/s did you make while solving this question?

The two genes assort independently (Mendel's law of independent assortment).

Q2. In a breed of dog, a dominant allele M is responsible for a merle coat pattern (mottled patches of colour). Dogs homozygous for the dominant allele (MM) are born with severe sensory defects and usually die just after birth. The recessive allele m produces a solid coat colour.

a) If two merle dogs (Mm) are crossed, what is the expected phenotypic ratio among the surviving offspring? Show your work using a Punnett square.

Punnett square:

	M	m
M	MM	Mm
m	Mm	mm

Genotypic ratio: 1 MM : 2 Mm : 1 mm

Phenotypes before mortality: 3 merle : 1 solid

MM dogs die, so among survivors:

Merle (Mm) : Solid (mm) = 2 : 1

b) A breeder crosses a merle dog with a solid-coloured dog and obtains 6 healthy puppies, all with merle coats. What is the genotype of the merle parent? Explain your reasoning.

Genotype: Mm

Reason: Since MM dies soon after birth, an adult merle dog must be Mm heterozygote. All offspring inherited M from the merle parent (Mm) and m from the solid parent (mm), giving Mm (merle). Given the small sample size, it is not unlikely that no mm progeny were produced, even though the cross was between Mm and mm individuals.

Q3. A geneticist is studying husk colour in barley grains. They perform the following crosses:

Cross 1: Pure-breeding white-husked barley $\times$ Pure-breeding black-husked barley $\rightarrow$ All F1 plants have black husks.

Cross 2: F1 black $\times$ F1 black $\rightarrow$ F2 shows 142 black, 36 white, and 12 brown barley plants.

a) Based on the F2 numbers (142:36:12), simplify this to the simplest whole number ratio. What genetic interaction does this ratio indicate?

Total F2 = 190.

Divide by 12: $142/12 \approx 12$, $36/12 = 3$, $12/12 = 1$.

Simplified ratio: 12 : 3 : 1

This indicates epistasis.

b) Using appropriate gene symbols, propose a genotype-to-phenotype model that explains these results. Assign genotypes to the original parents and the F1 generation.

A___ $\rightarrow$ black husk

aaB_ $\rightarrow$ white husk

aabb $\rightarrow$ brown husk

Parents:

Pure-breeding white: aaBB

Pure-breeding black: AAbb

F1: AaBb (all black)